Institute of Technology of Cambodia

Department of Information and Communication Engineering

Assignment: TP07

Subject: **Operating System**

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1. Read the “Part 4 - Writing Shell Scripts” in “The Linux Command Line 2nd Edition” by William E. Shotts, Jr, and answer the following questions:
   a. By using your my words, shell scripts are shell scripts are text files containing a series of commands written in a scripting language (like Bash) that the shell can interpret and execute. They automate tasks by allowing you to execute multiple commands sequentially, making complex tasks easier and more efficient.
   b. The format or header of a script file is `#!/bin/bash` (no extension .sh) or `#!bin/sh` (with extension .sh).
2. create your first script!
   a. Write your first script named as “hello_ubuntu” which displays “Hello! This is Ubuntu.”

```
sokmeak@sokmeaksaren:~/bin$ touch hello_ubuntu
```

   b. The command that I use to set permission for my script in order to make it executable for everyone is `chmod a+x hello_ubuntu`

```
sokmeak@sokmeaksaren:~/bin$ nano hello_ubuntu
sokmeak@sokmeaksaren:~/bin$ chmod a+x hello_ubuntu
sokmeak@sokmeaksaren:~/bin$ ls -l
total 24
drwxrwxr-x 2 sokmeak sokmeak 4096 Jan 11 07:23 Documents
-rwxrwxr-x 1 sokmeak sokmeak   43 Jan 11 08:06 hello_ubuntu
drwxrwxr-x 2 sokmeak sokmeak 4096 Jan 11 07:32 Others
```

   c. The command that you use to execute my script above is `./hello_ubuntu`
   d. Take a screenshot of content of the script and result from running it.

```
  GNU nano 7.2                                          hello_ubuntu
#!/bin/bash

echo "Hello! This is Ubuntu."
```

```
sokmeak@sokmeaksaren:~/bin$ nano hello_ubuntu
sokmeak@sokmeaksaren:~/bin$ chmod +x hello_ubuntu
sokmeak@sokmeaksaren:~/bin$ ./hello_ubuntu
Hello! This is Ubuntu.
sokmeak@sokmeaksaren:~/bin$
```

3. Let’s make your script easier to execute and more advanced!
   a. List down two options that makes my script executable by using this command hello_ubuntu.
      i. Move the script to a directory listed in the `PATH` environment variable, such as `/usr/local/bin`:

```
sokmeak@sokmeaksaren:~/bin$ sudo mv hello_ubuntu /usr/bin
[sudo] password for sokmeak:
Sorry, try again.
[sudo] password for sokmeak:
sokmeak@sokmeaksaren:~/bin$ hello_ubuntu
Hello! This is Ubuntu.
sokmeak@sokmeaksaren:~/bin$
```

ii. Add the current directory (.) to your PATH temporarily:

```
sokmeak@sokmeaksaren:~/bin$ export PATH=$PATH:$(pwd)
sokmeak@sokmeaksaren:~/bin$ hello_ubuntu
Hello! This is Ubuntu.
```

b. Command whoami displays the name of current user. Now, let's make a new script named as "hello_there" and commanded by hello_there which can recognize the current username, and it will display, for example, "Hello ratha! How are you today?", where "ratha" is the current user for this example

```
  GNU nano 7.2                                                    hello_there
#!/bin/bash

echo "Hello `whoami`! How are you today?"
```

```
sokmeak@sokmeaksaren:~/bin$ hello_there
Hello sokmeak! How are you today?
sokmeak@sokmeaksaren:~/bin$
```

4. Doing research on how to make a script run automatically whenever the terminal is opened, and then make your terminal run your last script "hello_there" every single time you open your terminal. After everything is set, take a screenshot to proof your success.

The key is add the bash script `hello_there` to the file `.bashrc` then it will run automatically whenever the terminal is opened.

```
sokmeak@sokmeaksaren: ~
  GNU nano 7.2                    /home/sokmeak/.bashrc
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi
export PATH=~/bin:"$PATH"
#getStarted
hello_there

^G Help      ^O Write Out  ^W Where Is   ^K Cut      ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste    ^J Justify    ^/ Go To Line
```

```
sokmeak@sokmeaksaren: ~
Hello sokmeak! How are you today?
sokmeak@sokmeaksaren:~$
```